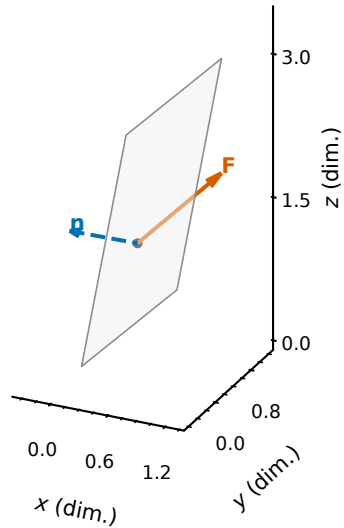
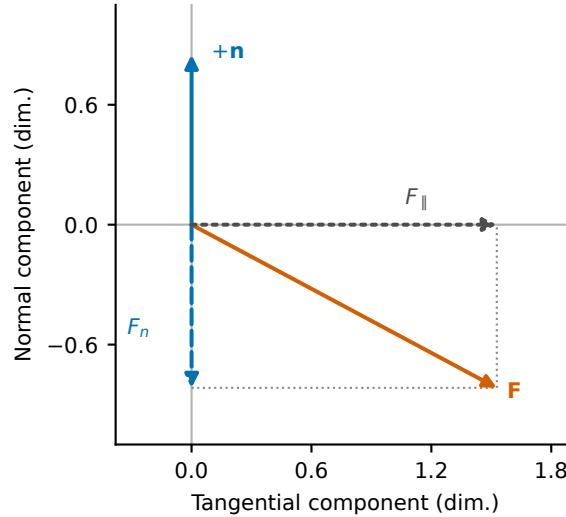
A



$\mathbf{n} = (-1, -2, 1)/\sqrt{6}$; $\mathbf{F} = (1, 1, 1)$

B



$\mathbf{F} \cdot \mathbf{n} = -2/\sqrt{6}$; $\mathrm{d}\Phi = -2\mathrm{d}u\,\mathrm{d}v$